Complete Classification of Full State Revivals on Separable Quantum Walks on Tori

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A quantum walk revives when it returns to its exact starting state after some number of steps. In this paper, we verify the work of Dukes on 1D cycles and extend it to toroidal lattices to find and classify all exact revivals in the 2D case. To achieve this, we use a separable coin parameterised by reflection parameters ρ_x, ρ_y and phase parameters δ_x, δ_y. In 2D, the separable walk factorises as a tensor product of two 1D walks. A torus revives when both axes revive. It can also revive when the two axes' phases cancel each other, at a period shorter than their least common multiple. The 2D classification contains revivals in 28 tori, 59 different periods, and 18,174 distinct parameter sets. On the 1D cycle, we establish within the search bound that exactly seven cycle lengths admit an exact revival in one dimension, $k \in \{2, 3, 4, 5, 6, 8, 10\}$ and verify the positive cases at the operator level. We run an operator-level search on a 1001×360 grid over (ρ, δ), for $k \leq 16$ and $N \leq 60$, and find no exact revival for the remaining cycle lengths. We find one correction to Dukes' 1D tables: the $k = 2$, $\rho = 1/2$, $\delta \in \{\pi/2, 3\pi/2\}$ exact revivals at $N = 24$ lie in Dukes' seed family but were not tabulated, because Dukes' original seeding was too constrained to reach them. We offer a conjecture, supported by a cyclotomic-field argument, that no cycle length other than $\{2, 3, 4, 5, 6, 8, 10\}$ admits an exact revival.

I. INTRODUCTION

Periodicity is a fundamental aspect of physics and a clock is an essential component of most computer systems. An analogous phenomenon occurs in the quantum realm: under certain conditions on symmetric topologies, a quantum walker returns to its initial state, both position and spin, after some number of steps. We call this a *full-state revival*. Such revivals have been proposed as a resource for quantum memory [1] and for quantum state routing [2], as a system that returns periodically to a previous state can serve as a clock, a register, or a routing primitive.

Here we consider a discrete-time quantum walk (DTQW) with a separable four-dimensional coin. For a DTQW, U is the single-step operator of the walk. A full-state revival at period N means $U^N = I$: the operator is the identity, so every state returns to itself with no phase after N steps. A related phenomenon also occurs along the walk: the state may return to itself up to an unobservable global phase, $U^N = e^{i\phi} I$ for some $\phi \neq 0$. We call this a *projective* revival. The classification in this paper is of full-state revivals in the strict sense, $U^N = I$; the projective revivals are noted where they occur but are not the object of classification.

Full-state revivals are rare. They occur only for certain coins on symmetric topologies, and their spectral identity is equally special: for $U^N = I$ to hold, every eigenphase θ_j of the walk operator U must be a rational multiple of 2π, $\theta_j = 2\pi m_j/N$ for integer m_j and a common period N. In other words, the eigenvalues of U must all be roots of unity, sharing a common power N that returns the walk to its initial state, $U^N = I$

Dukes [3] classified full-state revivals on 1D cycle graphs. Building on this work, we extend the classification to separable DTQWs over 2D toroidal lattices. A separable coin factorises into independent 1D coins on each axis of a $k \times m$ torus. The two factors revive independently, and the torus walk revives when they both do. The classification covers the admissible cycle lengths $k, m \in \{2, 3, 4, 5, 6, 8, 10\}$ with revival periods up to $N = 420$.

Dukes' coin is parameterised by a real reflection parameter ρ and a phase δ. The 2D coin has the corresponding parameters $\{\rho_x, \delta_x, \rho_y, \delta_y\}$. We report the classification as a set of records $(k, m, N, \rho_x, \delta_x, \rho_y, \delta_y)$, each record giving a torus, an exact revival period, and the coin parameters. The classification records each distinct parameter set at its smallest exact revival period.

Previous work on 2D lattices involved separable coins. Štefaňák et al. [4] stated that the four-state walk on a 2D lattice admits $N = 2$ revivals but no longer cycles within the class considered. Later Dodangodage [5] constructed walks with revivals of longer period and Jayakody [6] showed that a non-separable coin can be designed to produce revivals of any desired period. In that construction the period is a free parameter: the coin and shift are built so that the walk returns after any chosen number of steps. Our classification covers the separable case, where the (parameterised) coin and shift are fixed to the standard forms on each axis and the period is not chosen but emerges from the walk itself. We consider the two results to be complementary. Jayakody answers whether a revival of any given period can be built. More recently, Steinfurth et al. [7] demonstrated revivals experimentally using coupled optical-fibre loops. By changing the coin over time in a way that respects a simple mirror symmetry, their scheme can produce a revival at almost any chosen period for any input state, because the coin

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is modulated rather than fixed.

The remainder of this paper is structured as follows. Section II defines the separable walk and its four-parameter coin, gives a worked single-step example, and states the classification theorems. Section III describes how the classification was computed, and Section IV presents the results. Section V discusses the results and the classification, and Section VI summarises the output, states the open problem and the outlook.

II. THEORETICAL FRAMEWORK

A. Separable coin walk on a Torus

The single-step evolution of the walker on a DTQW is governed by the unitary operation

$$U = S\,(I_p \otimes C), \tag{1}$$

where I_p denotes the identity on the position space, C is the coin operator acting on the internal degree of freedom, and S is the conditional shift operator that entangles position and coin by displacing the walker according to its coin state.

In the 1D cycle setting, Dukes [3] introduced the parameterised coin operator

$$C(\rho,\delta) = \begin{pmatrix} \sqrt{\rho} & \sqrt{1-\rho}\,e^{i\delta} \\ \sqrt{1-\rho}\,e^{-i\delta} & -\sqrt{\rho} \end{pmatrix}, \tag{2}$$

where $\rho \in [0,1]$ physically dictates the spatial splitting ratio of the walker as it transitions to adjacent nodes, while $\delta \in [0, 2\pi)$ governs the relative phase accumulated between these divergent paths. We note that δ does not enter the trace of the single-step evolution operator for the 1D walk and consequently, the eigenphase spectrum—and thus the revival period—depends solely on ρ. The phase parameter δ merely redirects the walker along different directional paths without altering the revival periodicity. Setting $\rho = \frac{1}{2}$ and $\delta = 0$ creates the Hadamard coin:

$$C\left(\frac{1}{2},0\right) = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \equiv H. \tag{3}$$

The 1D construction extends to a two-dimensional $k \times m$ toroidal lattice. The total state space is assembled as a tensor product of position and coin Hilbert spaces, $\mathcal{H} = \mathcal{H}_p \otimes \mathcal{H}_c$. The spatial component $\mathcal{H}_p = \mathcal{H}_{p_x} \otimes \mathcal{H}_{p_y}$ is spanned by the basis $\{|x,y\rangle : x \in \mathbb{Z}_k, y \in \mathbb{Z}_m\}$, with periodic boundaries along both axes. The coin space $\mathcal{H}_c = \mathcal{H}_{c_x} \otimes \mathcal{H}_{c_y} \cong \mathbb{C}^4$ carries the directional basis $\{|L\rangle, |R\rangle\} \otimes \{|D\rangle, |U\rangle\}$, with L, R denoting displacements along the x axis and D, U along the y axis. The ordered basis is:

$$|LD\rangle, \quad |LU\rangle, \quad |RD\rangle, \quad |RU\rangle. \tag{4}$$

A separable coin factorises as $C_{k\times m} = C_x(\rho_x, \delta_x) \otimes C_y(\rho_y, \delta_y)$, with independent 1D parameters on each axis. In the ordered basis above, the matrix representation is

$$C_{k\times m} = \begin{pmatrix} \sqrt{\rho_x\rho_y} & \sqrt{\rho_x(1-\rho_y)}\,e^{i\delta_y} & \sqrt{(1-\rho_x)\rho_y}\,e^{i\delta_x} & \sqrt{(1-\rho_x)(1-\rho_y)}\,e^{i(\delta_x+\delta_y)} \\ \sqrt{\rho_x(1-\rho_y)}\,e^{-i\delta_y} & -\sqrt{\rho_x\rho_y} & \sqrt{(1-\rho_x)(1-\rho_y)}\,e^{i(\delta_x-\delta_y)} & -\sqrt{(1-\rho_x)\rho_y}\,e^{i\delta_x} \\ \sqrt{(1-\rho_x)\rho_y}\,e^{-i\delta_x} & \sqrt{(1-\rho_x)(1-\rho_y)}\,e^{-i(\delta_x-\delta_y)} & -\sqrt{\rho_x\rho_y} & -\sqrt{\rho_x(1-\rho_y)}\,e^{i\delta_y} \\ \sqrt{(1-\rho_x)(1-\rho_y)}\,e^{-i(\delta_x+\delta_y)} & -\sqrt{(1-\rho_x)\rho_y}\,e^{-i\delta_x} & -\sqrt{\rho_x(1-\rho_y)}\,e^{-i\delta_y} & \sqrt{\rho_x\rho_y} \end{pmatrix}. \tag{5}$$

Similar to the 1D case, the trace diagonal of this matrix contains only expressions in ρ_x and ρ_y, so the revival period depends only on these two parameters.

As the spatial axes are orthogonal and the coin acts separably, the conditional shift is $S = S_x S_y$. With subsystem identities $I_x \equiv I_{p_x} \otimes I_{c_x}$ and $I_y \equiv I_{p_y} \otimes I_{c_y}$, the two components are

$$S_x = \sum_{x\in\mathbb{Z}_k} \Big(|x-1\rangle\langle x| \otimes |L\rangle\langle L| + |x+1\rangle\langle x| \otimes |R\rangle\langle R| \Big) \otimes I_y, \tag{6}$$

$$S_y = I_x \otimes \sum_{y\in\mathbb{Z}_m} \Big(|y-1\rangle\langle y| \otimes |D\rangle\langle D| + |y+1\rangle\langle y| \otimes |U\rangle\langle U| \Big), \tag{7}$$

where all index shifts are understood modulo k and m, respectively (the toroidal boundary conditions).

The separability of the coin and the orthogonality of the axes yield a tensor-product factorisation of the full unitary. Reordering the total Hilbert space as $\mathcal{H} = (\mathcal{H}_{p_x} \otimes \mathcal{H}_{c_x}) \otimes (\mathcal{H}_{p_y} \otimes \mathcal{H}_{c_y})$, the single-step operator factorises as

$$U_{k\times m} = U_k \otimes U_m, \tag{8}$$

where $U_k = S_x(I_k \otimes C_x)$ and $U_m = S_y(I_m \otimes C_y)$ are independent 1D evolution operators on the x- and y-cycles. The factorisation isolates the spectral constraints along each axis, so the root-of-unity conditions that govern full-state revivals can be evaluated dimension-by-dimension.

A torus revival requires the sum of the two axis phases to be zero. When both axes revive at a common step, the sum vanishes at the LCM of the two axis periods; but two nonzero phases of opposite sign can also sum to zero at a step where neither axis has revived on its own, which gives a torus revival at a period shorter than the LCM of the two axis periods.

Note on exact versus projective revivals.

In all cases a revival requires both the position and spin state of the walker to return to its initial values. In this work we distinguish between two classes of revivals: a *projective* revival is one in which the state returns up to an unobservable global phase ($U^N = e^{i\phi} I$). An *exact* or full-state revival is one where the state returns with no phase ($U^N = I$). Here we enumerate only exact revivals.

B. The Mechanics of a Separable Walk on a Torus

To give an example of the separable walk, we evaluate a single step on a 3×3 torus with $\rho_x = \rho_y = 2/3$ and $\delta_x = \delta_y = 0$. The walker begins at the origin in coin state $|RU\rangle = |R\rangle \otimes |U\rangle$. The single-axis coin operator is

$$C(\tfrac{2}{3}, 0) = \begin{pmatrix} \sqrt{2/3} & \sqrt{1/3} \\ \sqrt{1/3} & -\sqrt{2/3} \end{pmatrix}, \tag{9}$$

acting on the basis $\{|\text{negative}\rangle, |\text{positive}\rangle\}$ along each axis. Its action on the initial state is

$$C_x|R\rangle = \sqrt{1/3}\,|L\rangle - \sqrt{2/3}\,|R\rangle, \tag{10}$$
$$C_y|U\rangle = \sqrt{1/3}\,|D\rangle - \sqrt{2/3}\,|U\rangle, \tag{11}$$

so that the tensor product $C_x \otimes C_y$ produces the post-coin superposition

$$(C_x \otimes C_y)|RU\rangle = \tfrac{1}{3}|LD\rangle - \tfrac{\sqrt{2}}{3}|LU\rangle - \tfrac{\sqrt{2}}{3}|RD\rangle + \tfrac{2}{3}|RU\rangle. \tag{12}$$

The conditional shift $S = S_x S_y$ then displaces each branch by one site in the direction indicated by the coin state, mapping the four basis states on the 3×3 torus as

$$|LD\rangle : (0,0) \to (2,2), \tag{13}$$
$$|LU\rangle : (0,0) \to (2,1), \tag{14}$$
$$|RD\rangle : (0,0) \to (1,2), \tag{15}$$
$$|RU\rangle : (0,0) \to (1,1). \tag{16}$$

The resulting state

$$|\psi_1\rangle = \tfrac{1}{3}|2,2\rangle|LD\rangle - \tfrac{\sqrt{2}}{3}|2,1\rangle|LU\rangle - \tfrac{\sqrt{2}}{3}|1,2\rangle|RD\rangle + \tfrac{2}{3}|1,1\rangle|RU\rangle, \tag{17}$$

is entangled in position⊗coin, with the norm preserved:

$$\langle\psi_1|\psi_1\rangle = \tfrac{1}{9} + \tfrac{2}{9} + \tfrac{2}{9} + \tfrac{4}{9} = 1. \tag{18}$$

Figure 1 shows the splitting of amplitude across the four directions and the subsequent distribution of the branches to their destination nodes.

Two features of this step are worth noting. First, position and coin are entangled after a single shift, reflecting the correlation between the two axes that the separable coin introduces. Second, unlike the $\rho = 1/2$ Hadamard case, the split is asymmetric: the $|RU\rangle$ branch retains amplitude $2/3$, the two off-diagonal branches carry $-\sqrt{2}/3$, and the opposite-corner branch carries $1/3$.

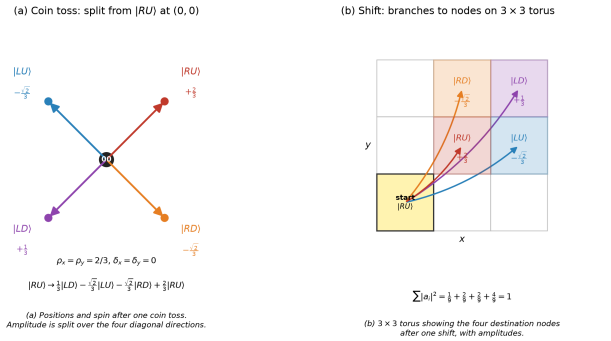


FIG. 1. One step of the separable walk on a 3×3 torus with $\rho_x = \rho_y = 2/3$ and $\delta_x = \delta_y = 0$, starting from the origin in coin state $|RU\rangle$. (a) The coin operator splits the amplitude over the four directions with weights $(1/3, -\sqrt{2}/3, -\sqrt{2}/3, 2/3)$ on the ordered basis $(|LD\rangle, |LU\rangle, |RD\rangle, |RU\rangle)$. (b) The conditional shift moves each branch to its destination node on the torus. The resulting state is entangled in position⊗coin.

C. Classification Theorems

Having established revivals on 1D cycles for $k \in \{8, 10\}$, one might intuitively anticipate that larger cycles—such as $k \in \{12, 16, \dots\}$—would similarly support full-state revivals with relative ease. Yet, as we shall demonstrate, this expectation proves incorrect: the algebraic constraints imposed by the dispersion relation become increasingly rigid as the cycle size grows.

Theorem 1 (1D existence)**.** *A discrete-time quantum walk on a 1D cycle graph $\mathbb{Z}_k$ driven by Dukes' coin $C(\rho,\delta) \in \mathrm{U}(2)$ admits a non-trivial full-state revival for every $k \in \{2,3,4,5,6,8,10\}$.*

Proof of Theorem 1. For $k \in \{2,3,4,6\}$ Dukes gives families of solutions [3], §4.2–4.4; for $k \in \{5,8,10\}$ he gives finite sets of explicit triples (ρ,δ,N) in §4.5. Each triple is a witness to the existence of a revival at that k. Substituting each witness into the dispersion relation $\sin\theta_j = \sqrt{\rho}\sin(2\pi j/k)$ and checking that every eigenphase is a rational multiple of $2\pi/N$ verifies $U_k^N = I_{2k}$ directly. □

Theorem 2 (1D non-existence within the search bound)**.** *An exhaustive search over a grid of $\rho \in \{j\cdot 10^{-4} : j = 1,\dots,9999\}$ and δ/π at $1/180$ spacing (360 base points), augmented by the Dukes-exact δ values, for revival step count $N \le 60$, finds no non-trivial full-state revival for any $k \notin \{2,3,4,5,6,8,10\}$ with $k \le 16$. The completeness of the 1D classification for $k \ge 17$ and $N > 60$ remains open (see Conjecture 3).*

Proof of Theorem 2. We performed a grid search over the parameter space $\rho \in \{j\cdot 10^{-4} : j = 1,\dots,9999\}$, δ/π at $1/180$ spacing (360 base points) augmented by the Dukes-exact δ values, and $N \in \{1,\dots,60\}$, for each $k \in \{2,\dots,16\}$. For every candidate, we evaluated the root-of-unity residual $r_N(\rho,\delta) = \max_j |\theta_j - m_j\cdot 2\pi/N|$. Grid points with residual below 10^{-3} were refined by bounded scalar minimisation followed by a local sweep, and accepted only if the residual after refinement fell below 10^{-10} and the independent matrix-exponentiation check $\|U_k^N - I_{2k}\| < 10^{-10}$ also passed. The search found no revival for any $k \in \{7,9,11,12,13,14,15,16\}$. The search is reproducible from the EXP032b notebook in the accompanying repository [8]. □

Corollary 2.1 (2D existence)**.** *A separable quantum walk on a $k\times m$ toroidal lattice exhibits a full-state revival whenever both spatial dimensions belong to the 1D revival set $k,m \in \{2,3,4,5,6,8,10\}$ and their revival period sets share a common element, $\mathcal{N}_k \cap \mathcal{N}_m \neq \emptyset$.*

Proof of Corollary 2.1. The 2D evolution operator factorises as $U_{k\times m} = U_k \otimes U_m$ following the tensor-product decomposition of Eq. (8). If $k,m \in \{2,3,4,5,6,8,10\}$, Theorem 1 supplies 1D witnesses. If $\mathcal{N}_k \cap \mathcal{N}_m \neq \emptyset$, choose a common period N. Then $U_{k\times m}^N = U_k^N \otimes U_m^N = e^{i(\phi_x+\phi_y)} I_{4km}$, the stated revival. □

Corollary 2.2 (2D non-existence within the search bound)**.** *For $k,m \le 16$, no $k\times m$ toroidal lattice outside the class of Corollary 2.1 exhibits a full-state revival at any period $N \le 60$.*

Proof of Corollary 2.2. Suppose a separable walk on a $k\times m$ torus admits a full-state revival at step N. By the factorisation $U_{k\times m} = U_k \otimes U_m$, this is equivalent to $U_k^N = e^{i\phi_x} I_{2k}$ and $U_m^N = e^{i\phi_y} I_{2m}$ simultaneously. By Theorem 2, no such $N \le 60$ exists when $k \notin \{2,3,4,5,6,8,10\}$ and $k \le 16$; the same holds for m. If both k and m are admissible but $\mathcal{N}_k \cap \mathcal{N}_m = \emptyset$, no common step count exists. □

Conjecture 3 (no revivals outside the admissible set)**.** *No cycle length $k \notin \{2,3,4,5,6,8,10\}$ admits a non-trivial 1D full-state revival at any period N or any coin parameter $\rho \in (0,1)$.*

Remark 1 (Evidence for Conjecture 3)**.** *The dispersion relation for a k-cycle is $\sin\theta_j = \sqrt{\rho}\,\sin(2\pi j/k)$ for $j = 1,\dots,k-1$. A revival at period N forces each eigenphase to satisfy $\theta_j = \pi r_j/N$ for some integer r_j. Taking the ratio of the dispersion relations for the two lowest spatial modes ($j=2$ and $j=1$) eliminates ρ entirely:*

$$\frac{\sin\theta_2}{\sin\theta_1} = \frac{\sin(4\pi/k)}{\sin(2\pi/k)} = 2\cos\left(\frac{2\pi}{k}\right). \tag{19}$$

The coin parameter $\sqrt{\rho}$ cancels out, leaving a purely geometric constant on the right-hand side. Because $\theta_1,\theta_2 \in \pi\mathbb{Q}$, that constant must equal a ratio of sines of rational angles. For $k=2$ the cycle has only one spatial harmonic ($j=1$), and the revival condition is trivially solvable by choice of ρ.

For $k \ge 3$ the constant $2\cos(2\pi/k)$ generates the real cyclotomic subfield $\mathbb{Q}(\cos(2\pi/k))$, whose degree over $\mathbb{Q}$ is $\phi(k)/2$, where ϕ is Euler's totient function [9]. The degree of this field therefore controls whether a single real parameter ρ can synchronise the harmonics. For $k \in \{3,4,5,6,8,10\}$ the degree is at most two, and the constant is rational or quadratic; for $k=12$ it is two; for $k \in \{7,9\}$ and $k \ge 11$ it is at least three.

The case $k=12$ also has $\phi(12)/2 = 2$, but the cycle carries three non-trivial harmonics ($j=1,2,3$). The third gives $\sin\theta_3 = \sqrt{\rho}\,\sin(\pi/2) = \sqrt{\rho}$. Niven's theorem restricts the sine of a rational angle to the values $0,\pm\frac{1}{2},\pm 1$ [10], so $\sqrt{\rho}$ must lie in that set. Alignment with the $j=1$ and $j=2$ harmonics then forces $\theta_3 = \pi/2$ uniquely, hence $\rho = 1$. The coin reduces to a deterministic reflector, and no non-trivial revival exists in the open interval $\rho \in (0,1)$.

For $k \in \{7,9\}$ and all $k \ge 11$ except $k = 12$, the degree satisfies $\phi(k)/2 \ge 3$. The minimal polynomial of $2\cos(2\pi/k)$ over $\mathbb{Q}$ then has degree at least three. If a revival existed, applying the automorphisms of $\mathrm{Gal}(\mathbb{Q}(\zeta_{2N})/\mathbb{Q})$ to the dispersion relations would map the trigonometric algebraic integers to their Galois conjugates, generating at least three independent constraints on the single parameter ρ. Whether these constraints are in fact independent is the open step. If they are, the system revives at only the trivial, boundary solutions $\rho \in \{0,1\}$.

We therefore state the conclusion as a conjecture, supported on the search bound of Theorem 2 and consistent with the explicit witnesses of Theorem 1.

III. METHODOLOGY

The classification is built and verified in two notebooks, `EXP033.ipynb` and `EXP032b.ipynb`, available in the accompanying repository [8]. The method proceeds in four stages.

Revival demonstration (EXP033). We evolve the walk step by step for $k = 4$, $\rho = 1/2$, $\delta = \pi/2$ and record the state at each step. The walker returns to its initial position and spin at $N \in \{6, 12, 18, 24\}$; these four revivals carry different global phases, three projective and one exact. This demonstrates the distinction between the two criteria on a single worked example.

1D verification (EXP033). We transcribe the complete set of 1D revival solutions tabulated by Dukes for cycle lengths $k \in \{2, 3, 4, 5, 6, 8, 10\}$, together with the $k = 2$, $\rho = 1/2$, $\delta \in \{\pi/2, 3\pi/2\}$ correction, and verify each entry at the operator level using the eigenvalue residual

$$r(k, \rho, \delta, N) = \max_j \left| \lambda_j^N - \overline{\lambda^N} \right|, \tag{20}$$

where λ_j are the eigenvalues of $U_k(\rho, \delta)$ and $\overline{\lambda^N}$ is their N-th power mean. The residual vanishes when $U^N = e^{i\phi} I$; the exact case $U^N = I$ is confirmed separately by a matrix-power check at the recorded period.

Exhaustive grid search (EXP032b). We search for exact revivals on 1D cycles for $k \leq 16$ on a 1001×360 grid over $(\rho, \delta/\pi)$, with ρ at 10^{-3} spacing and δ/π at $1/180$ spacing, augmented by the Dukes-exact δ values, and test $U^N = I$ up to $N = 60$ at each grid point. The operator is diagonalised once per grid point, and the eigenvalue residual $\max_j |\lambda_j^N - 1|$ is evaluated for every N. Grid points that pass the tolerance are recorded as *candidate revivals*. Each candidate is then refined: its parameters are snapped to the nearest algebraic value, Snapping moves the point by at most 10^{-3} in ρ and 10^{-3} in δ/π, and the snapped point must then pass a stricter check than the detection threshold: 10^{-12} on the eigenvalue residual and 10^{-10} on the matrix-power residual, against the 10^{-9} used for detection. Candidates that fail refinement are reported as `unconfirmed` and treated as a bug rather than a result. the exact period is confirmed by a direct matrix-power check $\|U^N - I\| < 10^{-10}$, and the confirmed revival is classified as trivial, Dukes-tabulated, or new. Trivial revivals are those at the degenerate parameter values $\rho \in \{0, 1\}$ or $\delta \in \{0, \pi\}$.

2D enumeration (EXP033). For a separable walk on a $k \times m$ torus, the single-step operator factorises as $U_{k\times m} = U_k \otimes U_m$. We enumerate all ordered pairs (k, m) with $k, m \in \{2, 3, 4, 5, 6, 8, 10\}$ and $k \leq m$, every pair of 1D signatures on those cycles, and every N from 1 to 420. For each pair, the operator is diagonalised once and the projective residual is evaluated at every N; the exact period is derived from the phase order via $N_{\text{exact}} = N_{\text{proj}} \cdot \text{ord}(\phi)$, where $\text{ord}(\phi)$ is the smallest r with $\phi^r = 1$. Scanning every N rather than only multiples of the factor LCM is necessary because the phase-cancellation case can revive exactly at a proper divisor of the LCM. Each canonical record is confirmed by a matrix-power check $\|U^N - I\| < 10^{-10}$ at its exact period.

IV. EXPERIMENTAL RESULTS

Demonstration. Table I shows the step-by-step amplitudes at the projective and exact revival steps for this $k = 4$ case. While the walker's position and spin return to the initial state at $N = 6, N = 12, N = 18$ and $N = 24$, an exact revival only occurs at $N = 24$ when the initial phase, $e^{i\phi} = 1$. The revivals at $N = 6, 12, 18$ are called projective revivals as $e^{i\phi} \neq 1$ at these steps.

1D catalogue. The catalogue contains 172 signatures across the seven cycle lengths $k \in \{2, 3, 4, 5, 6, 8, 10\}$, transcribed from Dukes' tables together with the $k = 2$ correction. Every signature is verified at the operator level, and the verification pass is reported in Table V in the Appendix,

1D exhaustive grid search. On a 1001×360 grid over (ρ, δ), for $k \leq 16$ and $N \leq 60$, the exhaustive search finds no exact revival for $k \notin \{2, 3, 4, 5, 6, 8, 10\}$ outside the trivial revivals at $\rho \in \{0, 1\}$ and $\delta \in \{0, \pi\}$. Every candidate recorded by the grid search is confirmed by a matrix-power check at its exact period.

2D enumeration. The 2D stage, built from the tensor factorisation, produces revivals on 28 distinct tori, at 59 distinct periods, and with a total of 18,174 distinct parameter sets. Each parameter set is recorded once, at its smallest exact revival period; the classification is available as supplementary data. Table II lists the 28 tori with their record counts and period ranges.

TABLE I. Step-by-step amplitudes at the revival steps for the discrete-time quantum walk on a 4-cycle with $\rho = 1/2$ and $\delta = \pi/2$, starting from $|0, \uparrow\rangle$. Only the steps at which the operator satisfies $U^N = e^{i\phi} I$ are shown ($N = 6, 12, 18, 24$), together with the initial state at $N = 0$. At each shown step the walker is localised at site 0; amplitudes on sites 1, 2, 3 are zero and are omitted. The phase column gives $e^{i\phi}$ in $U^N = e^{i\phi} I$. Rows marked † are projective revivals; the row marked †‡ is the unique exact revival, at which $e^{i\phi} = 1$ and $U^N = I$.

N	spin up		spin down		$e^{i\phi}$	tag
	a	ib	a	ib		
0	1	0	0	0	1	
6	0	1	0	0	$+i$	†
12	-1	0	0	0	-1	†
18	0	-1	0	0	$-i$	†
24	1	0	0	0	$+1$	†‡

† projective revival: $U^N = e^{i\phi} I$ with $e^{i\phi} \neq 1$. ‡ exact revival: $U^N = I$, i.e. $e^{i\phi} = 1$.

Period structure. Table III shows the revival periods observed on each torus, for $N \leq 60$ and $N = 120$. A bul-

TABLE II. Per-torus summary: reduced 2D revivals and range of revival periods. The table is split into three blocks; read the left block first, then the middle, then the right.

Torus	Records	N range	max N	Torus	Records	N range	max N	Torus	Records	N range	max N
2×2	9	2–24	24	3×6	1517	8–420	420	5×10	100	60–120	120
2×3	123	8–312	312	3×8	164	24–312	312	6×6	1369	8–420	420
2×4	201	6–312	312	3×10	340	60–420	420	6×8	148	24–312	312
2×5	30	60–120	120	4×4	4489	6–420	420	6×10	300	60–420	420
2×6	111	8–312	312	4×5	560	60–420	420	8×8	16	24–24	24
2×8	12	24–24	24	4×6	2479	8–420	420	8×10	40	120–120	120
2×10	30	60–120	120	4×8	268	24–312	312	10×10	100	60–60	60
3×3	1681	8–420	420	4×10	560	60–420	420				
3×4	2747	8–420	420	5×5	100	60–60	60				
3×5	340	60–420	420	5×6	300	60–420	420				
				5×8	40	120–120	120				

let marks each (k, m) and N where at least one distinct revival exists. The distribution is highly uneven. The tori with a $k = 2$ axis contribute the smallest families (Table II): the 2-cycle has only three signatures, at minimal periods 2, 24, and 24, and the resulting torus families are correspondingly sparse. The largest families occur on the tori 4×4, 3×4, and 3×3, where the dense 1D spectra of the 3- and 4-cycles allow many parameter combinations to share a common period. The period $N = 60$ appears on the most tori, because the 5- and 10-cycle signatures all have minimal period 60, which dominates the LCM with any admissible axis period; every torus with a 5 or 10 axis therefore inherits that period. The 5×8 and the 8×8 tori have their first revival at $N = 120$.

The 59 revival periods range from $N = 2$ to the scan bound $N = 420$. The largest counts occur at the highly composite periods, notably $N = 60$ with 2,456 records, $N = 120$ with 1,642, $N = 420$ with 1,076, $N = 24$ with 719, and $N = 30$ with 581. These values are the LCMs of many pairs of 1D revival periods and therefore support many parameter combinations. The smallest periods are sparse: $N = 2$ appears only on the 2×2 torus at $\delta_x = \delta_y = 0$, while $N = 6, 8, 10$ appear on a handful of tori each.

Phase selection. Table IV records the distribution of the phase parameters across the 18,174 distinct revivals. In 1D, the phase δ selects which (ρ, δ) pairs lie in the catalogue but does not change the period at which a given ρ revives, once it is in the catalogue; the sole exception is the $k = 2$ case, where $\rho = 1/2$ revives at $N = 2$ for $\delta = 0$ and at $N = 24$ for $\delta = \pi/2$ or $3\pi/2$. In 2D, by contrast, δ becomes a selection mechanism in its own right. Only 25.9% of the classification has $\delta_x = \delta_y = 0$. A further 48.9% has zero phase on exactly one axis (either $\delta_x = 0$ or $\delta_y = 0$, but not both), 5.0% has $\delta_x = \delta_y \neq 0$, and the remaining 20.2% has unequal non-zero phases on the two axes. All appearing phases are rational multiples of π with denominators dividing $2k$. The full 1D signature catalogue is given in Table V in the Appendix.

V. DISCUSSION

A. 2D Classification

The results of the tensor factorisation show that: 28 tori revive, at 59 distinct periods, with 18,174 distinct parameter sets in total. The distribution is highly uneven. The largest families occur on the tori 4×4, 3×4, and 3×3, which together account for just under half the classification. These are the cycles with the densest 1D spectra: the 3-cycle carries 41 signatures and the 4-cycle 69, so their tensor products generate many distinct parameter combinations. The tori involving $k = 2$ are the sparsest, because the 2-cycle carries only three signatures, at $N_{\min} \in \{2, 24, 24\}$.

Note the 1D grid search stops at $N \leq 60$, while the 2D classification extends to $N \leq 420$. The 2D enumeration is built from the catalogued 1D signatures, whose periods are all at most 60, and a 2D period is either one of those periods or a combination of a pair of them. The 2D scan therefore inherits the 1D catalogue rather than researching 1D at larger N; periods above 60 arise only as combinations of 1D revival periods.

The period spectrum is also scattered. Of the 59 distinct periods, the majority are populated by only a handful of parameter sets. The peak periods are $N = 60$ with 2,456 distinct parameter sets, $N = 120$ with 1,642, $N = 420$ with 1,076, $N = 24$ with 719, and $N = 30$ with 581. Each is highly composite and hence an LCM of many pairs of 1D revival periods, so a large number of 2D combinations share it. At the other extreme, the periods $N = 2$ and $N = 6$ are isolated on one torus and on three tori respectively.

The reflection (ρ_x, ρ_y) parameters take several algebraic forms. The low-lying signatures are rational ($1/2$, $1/3$, $1/4$, $2/3$, $3/4$) or quadratic surds of the form $(a \pm \sqrt{b})/c$ with $b \in \{2, 3, 5\}$. The surds split into three families: the $\sqrt{2}$-family, associated with the 4-cycle; the $\sqrt{3}$-family, associated with the 3- and 6-cycles; and the pentagonal family $(5 \pm \sqrt{5})/10$, associated with the 5- and 10-cycles. The higher-period signatures at $N \geq 14$ on the 3- and

TABLE III. Revival periods observed on each torus for $N \leq 60$ and $N = 120$. A bullet indicates at least one distinct revival at that period.

Torus	2	6	8	10	12	14	16	18	20	22	24	26	28	30	36	40	42	48	56	60	⋯	120
2×2	•	•			•						•											
2×3			•	•	•	•	•	•	•	•	•	•	•	•				•				•
2×4		•	•	•	•	•	•	•	•	•	•	•	•	•				•				•
2×5																				•		•
2×6			•	•	•	•	•	•	•	•	•	•	•	•				•				•
2×8											•											
2×10																				•		•
3×3			•	•	•	•	•	•	•	•	•	•	•	•	•	•		•	•	•		•
3×4			•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•		•
3×5																				•		•
3×6			•	•	•	•	•	•	•	•	•	•	•	•	•	•		•	•	•		•
3×8											•							•				•
3×10																				•		•
4×4		•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•		•
4×5																				•		•
4×6			•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•		•
4×8											•							•				•
4×10																				•		•
5×5																				•		
5×6																				•		•
5×8																						•
5×10																				•		•
6×6			•	•	•	•	•	•	•	•	•	•	•	•	•	•		•	•	•		•
6×8											•							•				•
6×10																				•		•
8×8											•											
8×10																						•
10×10																				•		

TABLE IV. Distribution of phase parameters across the reduced 2D revivals.

Category	Count	Percentage
$\delta_x = \delta_y = 0$	4710	25.9%
exactly one of δ_x, δ_y zero	8891	48.9%
$\delta_x = \delta_y \neq 0$	907	5.0%
$\delta_x \neq \delta_y$, both non-zero	3666	20.2%
Total	18174	100%

4-cycles are trigonometric rather than quadratic — for example the $k = 3$ value 0.251007 at $N = 14$ and the $k = 4$ value 0.188255 at $N = 14$ — but they remain algebraic and their minimal polynomials are determined by the corresponding root-of-unity angle. The duality $\rho \leftrightarrow 1 - \rho$ holds without exception (due to the definition of the coin), so each reflection parameter is paired with its complement elsewhere in the classification.

The phase (δ_x, δ_y) parameters are also a notable feature of the 2D classification. In the 1D case, δ affects only the eigenvectors of U_k; the eigenvalues, and hence the revival period, depend solely on ρ. This simplicity does not survive the tensor product. In 2D, the eigenvalues of $U_{k\times m}$ are products $\lambda_i \mu_j$, and a period-N revival requires the phases of the two factors to be compatible. The phase parameters δ_x and δ_y control which pairs of 1D eigenbases can be aligned. As Table IV shows, only 25.9% of the classification has $\delta_x = \delta_y = 0$, while 48.9% has zero phase on exactly one axis, 5.0% has $\delta_x = \delta_y \neq 0$, and the remaining 20.2% requires unequal non-zero phases on the two axes. All appearing phases are rational multiples of π whose denominators divide twice the cycle length. The phase parameters therefore act as a selection mechanism: the great majority of 2D revivals require a non-zero phase on at least one axis.

B. The 10×10 revival: a case study

The 10×10 torus is the largest torus admitting the period $N = 60$, and provides a clean illustration of the revival mechanism. We take $\rho_x = \rho_y = (5 - \sqrt{5})/10 \approx 0.2764$ and $\delta_x = \delta_y = 0$.

Figure 3 tracks the fidelity of the walker against the initial state over eighty steps. The fidelity reaches 1 at $N = 60$ No other step in the range produces a full-state revival. Figure 2 shows the position probability distribution at ten snapshots. The walker starts localised at the origin, spreads outward over the first thirty steps, and refocuses at $N = 60$.

Figure 4 shows the four components of the spin dis-

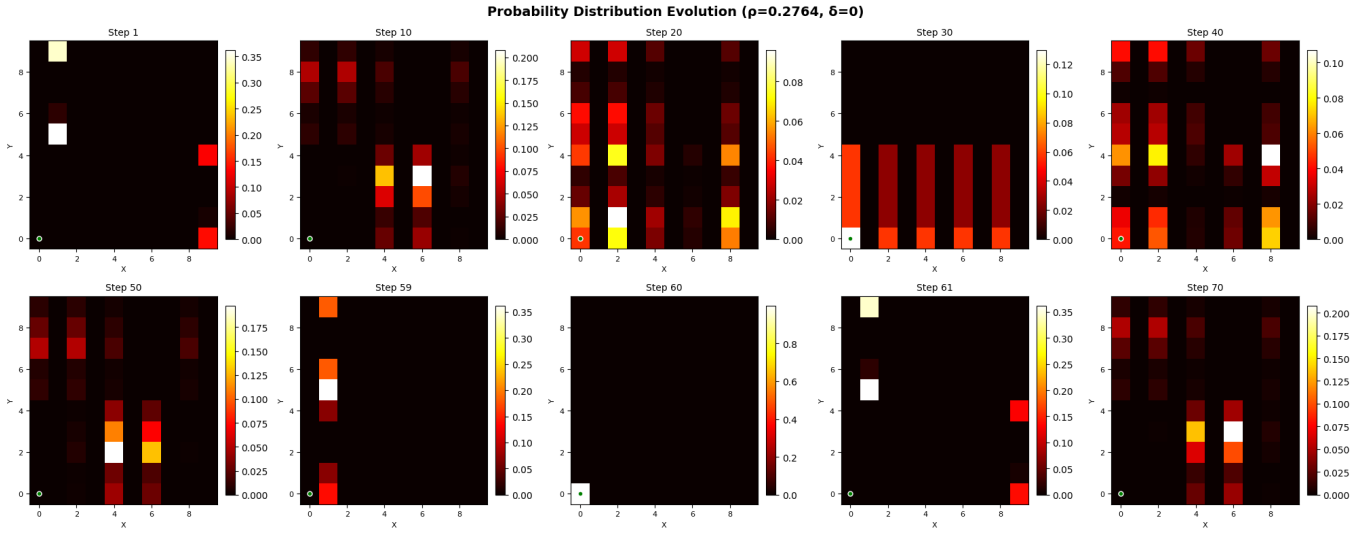


FIG. 2. Position probability distribution at ten snapshots between step 1 and step 70, for the same parameters as Fig. 3. The walker starts localised at the origin, spreads outward over the first thirty steps, and refocuses to the origin at $N = 60$. The distribution at step 60 matches the initial condition; the distributions at intermediate steps do not.

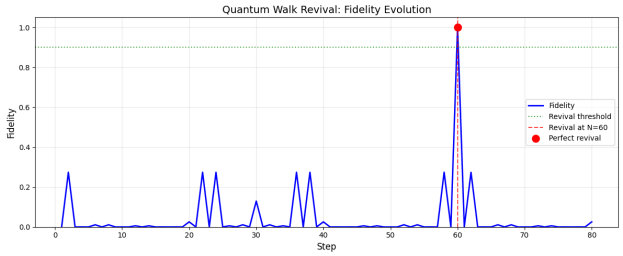


FIG. 3. Fidelity of the joint position⊗coin state against the initial state, for the 10×10 torus with $\rho_x = \rho_y = (5-\sqrt{5})/10 \approx 0.2764$ and $\delta_x = \delta_y = 0$. The fidelity reaches 1 at $N = 60$, confirming the exact full-state revival. A partial peak appears at $N = 30$, but the state does not fully refocus at that step.

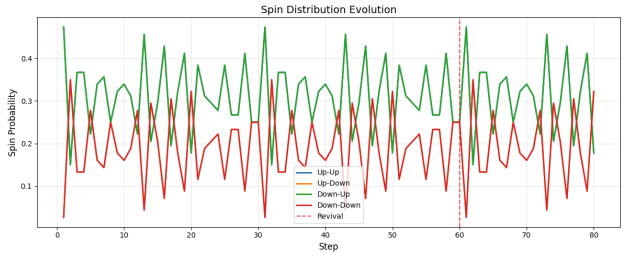


FIG. 4. Spin (coin) distribution over eighty steps for the 10×10 torus at $\rho = (5-\sqrt{5})/10$, $\delta = 0$. The four coin components return to their initial values at both $N = 30$ and $N = 60$. The return at $N = 30$ is a partial revival of the coin alone; only at $N = 60$ does the position distribution also coincide with the initial condition, producing the full-state revival.

tribution. The coin state returns to its initial values at both $N = 30$ and $N = 60$. The return at $N = 30$ is a *partial* revival: the coin reconstructs while the position distribution has not yet refocused. Only at $N = 60$ do coin and position both coincide with their initial values. The distinction is visible directly in Figs. 3 and 4.

VI. CONCLUSION

We have presented a complete classification of full-state revivals for separable quantum walks on $k \times m$ toroidal lattices, within the search bounds $k, m \in \{2, 3, 4, 5, 6, 8, 10\}$ and $N \leq 420$. The 1D case is exhaustive on the search grid: exactly seven cycle lengths admit revivals, and the operator-level verification of all 172 signatures is reported in Table V in the Appendix. The 2D case factorises via the tensor product, giving 28 tori, 59 periods, and 18,174 distinct parameter sets. We demonstrated an example of exact versus projective revivals in 1D, reported a single correction to Dukes' 1D tables — the $k = 2$, $\rho = 1/2$, $\delta \in \{\pi/2, 3\pi/2\}$ exact revival at $N = 24$, which lies in Dukes' seed family but was not tabulated — and stated a conjecture that the 1D admissible set is complete beyond the search bound, sketching a cyclotomic-field approach to the proof.

Summarising the 2D classification, two features stand out. First, the phase parameters δ_x and δ_y are a selection mechanism in 2D: only 25.9% of the distinct classification has $\delta_x = \delta_y = 0$, 48.9% has zero phase on exactly one axis, 5.0% has $\delta_x = \delta_y \neq 0$, and the remaining 20.2% has unequal non-zero phases on the two axes; thus 74.1% of the classification requires a non-zero phase on at least one axis. Second, the period spectrum is dominated by the highly composite periods: $N = 120$ occurs on 22

tori, $N = 60$ on 25, and $N = 24$ on 9, while the sparsest periods ($N = 2$, $N = 42$) reach only one and three tori respectively. These results open directions for quantum state routing and quantum memory architectures, and invite a proof of the stated conjecture.

USE OF AI-ASSISTED TOOLS

The core mathematics and analysis programs for the 1D cycle and 2D torus classifications were developed by the authors. Once that foundation was trusted and verified, the authors used DeepSeek (chat interface; model version not exposed at time of use - Aug-Oct 2026) to refine and augment the work into the final versions of the notebooks, `EXP032b.ipynb`, `EXP033.ipynb`. The AI assisted in three further areas: developing theorems: the cyclotomic-field sketch of Conjecture 3; migrating the numerical results from the notebook outputs into the LaTeX tables; and producing the fidelity and distribution plots that accompany the notebook. It was used to review drafts of the manuscript text for correctness.

The AI-assisted output was checked at each stage. The notebook verifies every 1D signature and every 2D record against an operator-level eigenvalue residual, and the results were compared with the tabulated values of Dukes [3]. All tables were regenerated from the notebook. The authors reviewed the AI-assisted output and checked the tables against the classification data. They take full responsibility for the accuracy, originality, and integrity of the manuscript.

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VII. APPENDIX

We collect in this appendix the complete catalogue of 1D signatures.

TABLE V: Full list of 1D full-state revivals, with reflection parameters in exact form.

k	ρ	δ	$N_{\min}$	source
2	$1/2$	0	2	Dukes T2 (continuous)
2	$1/2$	$\pi/2$	24	new
2	$1/2$	$3\pi/2$	24	new
3	$2/3$	0	8	Dukes T3
3	$\frac{5-\sqrt{5}}{6}$	0	10	Dukes T3
3	$1/3$	0	12	Dukes T3
3	$1/3$	$2\pi/3$	12	Dukes T3
3	$1/3$	$4\pi/3$	12	Dukes T3
3	0.251007	0	14	Dukes T3
3	0.815014	0	14	Dukes T3
3	0.195262	0	16	Dukes T3
3	0.155970	0	18	Dukes T3
3	0.550901	0	18	Dukes T3
3	0.155970	$2\pi/3$	18	Dukes T3
3	0.550901	$2\pi/3$	18	Dukes T3
3	0.155970	$4\pi/3$	18	Dukes T3
3	0.550901	$4\pi/3$	18	Dukes T3
3	0.127322	0	20	Dukes T3
3	0.872678	0	20	Dukes T3
3	0.105831	0	22	Dukes T3
3	0.389723	0	22	Dukes T3
3	0.761543	0	22	Dukes T3
3	0.089316	0	24	Dukes T3
3	0.089316	$2\pi/3$	24	Dukes T3
3	0.089316	$4\pi/3$	24	Dukes T3
3	$2/3$	$2\pi/3$	24	Dukes T3
3	$2/3$	$4\pi/3$	24	Dukes T3
3	0.076363	0	26	Dukes T3
3	0.287957	0	26	Dukes T3
3	0.586309	0	26	Dukes T3
3	0.903070	0	26	Dukes T3
3	0.066021	0	28	Dukes T3
3	0.518319	0	28	Dukes T3
3	0.057636	0	30	Dukes T3
3	0.220580	0	30	Dukes T3
3	0.736352	0	30	Dukes T3
3	0.057636	$2\pi/3$	30	Dukes T3
3	0.220580	$2\pi/3$	30	Dukes T3
3	0.736352	$2\pi/3$	30	Dukes T3
3	0.057636	$4\pi/3$	30	Dukes T3
3	0.220580	$4\pi/3$	30	Dukes T3
3	0.736352	$4\pi/3$	30	Dukes T3
3	$\frac{5-\sqrt{5}}{6}$	$2\pi/3$	30	Dukes T3
3	$\frac{5-\sqrt{5}}{6}$	$4\pi/3$	30	Dukes T3
4	$3/4$	0	6	Dukes T4
4	$1/2$	0	8	Dukes T4
4	$1/2$	π	8	Dukes T4
4	$\frac{5-\sqrt{5}}{8}$	0	10	Dukes T4
4	$\frac{5+\sqrt{5}}{8}$	0	10	Dukes T4
4	$1/4$	0	12	Dukes T4
4	$1/4$	π	12	Dukes T4
4	$3/4$	π	12	Dukes T4
4	$\frac{2-\sqrt{3}}{2}$	$\pi/2$	12	Dukes T4
4	$\frac{2-\sqrt{3}}{2}$	$3\pi/2$	12	Dukes T4
4	0.188255	0	14	Dukes T4
4	0.611260	0	14	Dukes T4
4	0.950484	0	14	Dukes T4
4	0.146447	0	16	Dukes T4
4	0.146447	π	16	Dukes T4
4	0.853553	0	16	Dukes T4
4	0.853553	π	16	Dukes T4
4	0.292893	$\pi/2$	16	Dukes T4
4	0.292893	$3\pi/2$	16	Dukes T4
4	0.116978	0	18	Dukes T4
4	0.413176	0	18	Dukes T4
4	0.095492	0	20	Dukes T4
4	0.095492	π	20	Dukes T4
4	0.654509	0	20	Dukes T4
4	0.654509	π	20	Dukes T4
4	$\frac{5-\sqrt{5}}{8}$	π	20	Dukes T4
4	$\frac{5+\sqrt{5}}{8}$	π	20	Dukes T4
4	0.048943	$\pi/2$	20	Dukes T4
4	0.412215	$\pi/2$	20	Dukes T4
4	0.048943	$3\pi/2$	20	Dukes T4
4	0.412215	$3\pi/2$	20	Dukes T4
4	0.079373	0	22	Dukes T4
4	0.292292	0	22	Dukes T4
4	0.571157	0	22	Dukes T4
4	0.827430	0	22	Dukes T4
4	0.979746	0	22	Dukes T4
4	0.066987	0	24	Dukes T4
4	0.066987	π	24	Dukes T4
4	0.933013	0	24	Dukes T4
4	0.933013	π	24	Dukes T4
4	$1/2$	$\pi/2$	24	Dukes T4
4	$1/2$	$3\pi/2$	24	Dukes T4
4	0.057272	0	26	Dukes T4
4	0.215968	0	26	Dukes T4
4	0.439732	0	26	Dukes T4
4	0.677302	0	26	Dukes T4
4	0.874255	0	26	Dukes T4
4	0.985471	0	26	Dukes T4
4	0.049516	0	28	Dukes T4
4	0.049516	π	28	Dukes T4
4	0.388740	0	28	Dukes T4
4	0.388740	π	28	Dukes T4
4	0.811745	0	28	Dukes T4
4	0.811745	π	28	Dukes T4
4	0.950484	π	28	Dukes T4
4	0.611260	π	28	Dukes T4
4	0.188255	π	28	Dukes T4
4	0.566116	$\pi/2$	28	Dukes T4
4	0.025072	$\pi/2$	28	Dukes T4
4	0.218169	$\pi/2$	28	Dukes T4
4	0.566116	$3\pi/2$	28	Dukes T4
4	0.025072	$3\pi/2$	28	Dukes T4
4	0.218169	$3\pi/2$	28	Dukes T4
4	0.043227	0	30	Dukes T4
4	0.165435	0	30	Dukes T4
4	0.552264	0	30	Dukes T4
4	0.989074	0	30	Dukes T4
5	$\frac{5-\sqrt{5}}{10}$	0	60	Dukes T5
5	$\frac{5+\sqrt{5}}{10}$	0	60	Dukes T5
5	$\frac{5-\sqrt{5}}{10}$	$2\pi/5$	60	Dukes T5
5	$\frac{5+\sqrt{5}}{10}$	$2\pi/5$	60	Dukes T5
5	$\frac{5-\sqrt{5}}{10}$	$4\pi/5$	60	Dukes T5
5	$\frac{5+\sqrt{5}}{10}$	$4\pi/5$	60	Dukes T5
5	$\frac{5-\sqrt{5}}{10}$	$6\pi/5$	60	Dukes T5

k	ρ	δ	$N_{\min}$	source
5	$\frac{5+\sqrt{5}}{10}$	$6\pi/5$	60	Dukes T5
5	$\frac{5-\sqrt{5}}{10}$	$8\pi/5$	60	Dukes T5
5	$\frac{5+\sqrt{5}}{10}$	$8\pi/5$	60	Dukes T5
10	$\frac{5-\sqrt{5}}{10}$	0	60	Dukes T5
10	$\frac{5+\sqrt{5}}{10}$	0	60	Dukes T5
10	$\frac{5-\sqrt{5}}{10}$	$2\pi/5$	60	Dukes T5
10	$\frac{5+\sqrt{5}}{10}$	$2\pi/5$	60	Dukes T5
10	$\frac{5-\sqrt{5}}{10}$	$4\pi/5$	60	Dukes T5
10	$\frac{5+\sqrt{5}}{10}$	$4\pi/5$	60	Dukes T5
10	$\frac{5-\sqrt{5}}{10}$	$6\pi/5$	60	Dukes T5
10	$\frac{5+\sqrt{5}}{10}$	$6\pi/5$	60	Dukes T5
10	$\frac{5-\sqrt{5}}{10}$	$8\pi/5$	60	Dukes T5
10	$\frac{5+\sqrt{5}}{10}$	$8\pi/5$	60	Dukes T5
6	$2/3$	0	8	Dukes T3 (k=6)
6	$\frac{5-\sqrt{5}}{6}$	0	10	Dukes T3 (k=6)
6	$1/3$	0	12	Dukes T3 (k=6)
6	$1/3$	$2\pi/3$	12	Dukes T3 (k=6)
6	$1/3$	$4\pi/3$	12	Dukes T3 (k=6)
6	0.251007	0	14	Dukes T3 (k=6)
6	0.815014	0	14	Dukes T3 (k=6)
6	0.195262	0	16	Dukes T3 (k=6)
6	0.155970	0	18	Dukes T3 (k=6)
6	0.155970	$2\pi/3$	18	Dukes T3 (k=6)
6	0.155970	$4\pi/3$	18	Dukes T3 (k=6)
6	0.550901	0	18	Dukes T3 (k=6)
6	0.550901	$2\pi/3$	18	Dukes T3 (k=6)
6	0.550901	$4\pi/3$	18	Dukes T3 (k=6)
6	0.127322	0	20	Dukes T3 (k=6)
6	0.872678	0	20	Dukes T3 (k=6)
6	0.105831	0	22	Dukes T3 (k=6)
6	0.389723	0	22	Dukes T3 (k=6)
6	0.761543	0	22	Dukes T3 (k=6)
6	0.089316	0	24	Dukes T3 (k=6)
6	0.089316	$2\pi/3$	24	Dukes T3 (k=6)
6	0.089316	$4\pi/3$	24	Dukes T3 (k=6)
6	0.076363	0	26	Dukes T3 (k=6)
6	0.287957	0	26	Dukes T3 (k=6)
6	0.586309	0	26	Dukes T3 (k=6)
6	0.903070	0	26	Dukes T3 (k=6)
6	0.066021	0	28	Dukes T3 (k=6)
6	0.518319	0	28	Dukes T3 (k=6)
6	0.057636	0	30	Dukes T3 (k=6)
6	0.057636	$2\pi/3$	30	Dukes T3 (k=6)
6	0.057636	$4\pi/3$	30	Dukes T3 (k=6)
6	0.220580	0	30	Dukes T3 (k=6)
6	0.220580	$2\pi/3$	30	Dukes T3 (k=6)
6	0.220580	$4\pi/3$	30	Dukes T3 (k=6)
6	0.736352	0	30	Dukes T3 (k=6)
6	0.736352	$2\pi/3$	30	Dukes T3 (k=6)
6	0.736352	$4\pi/3$	30	Dukes T3 (k=6)
8	$1/2$	0	24	Dukes T5 (k=8)
8	$1/2$	$\pi/2$	24	Dukes T5 (k=8)
8	$1/2$	π	24	Dukes T5 (k=8)
8	$1/2$	$3\pi/2$	24	Dukes T5 (k=8)

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